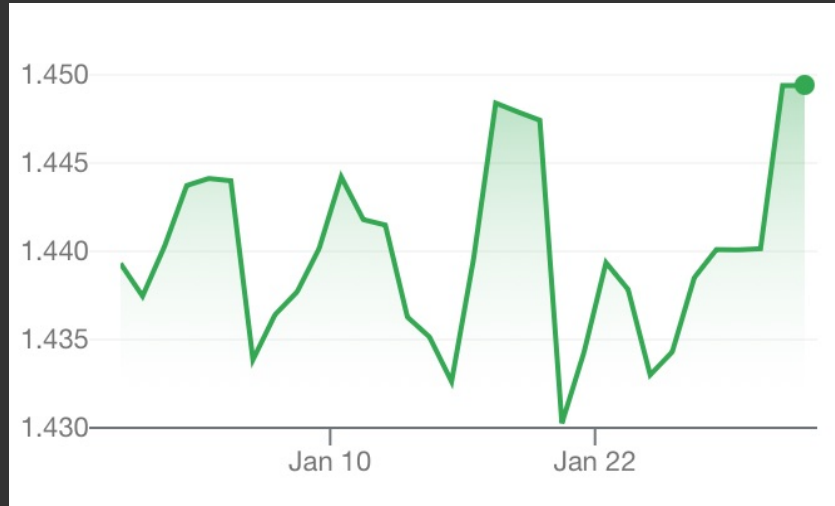


variables

CSCI
134

as of late
january



$$= 1.45 *$$

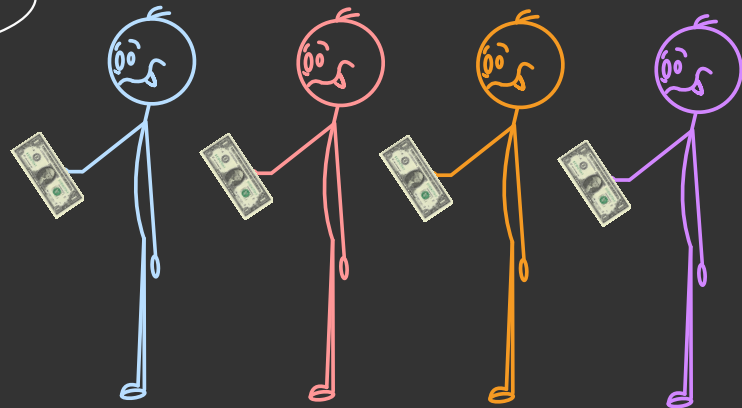


one **usa dollar** was worth 1.45 **canadian dollars**

I want to exchange
\$15 for canadian money

here's **\$21.75**

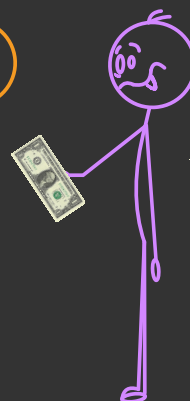
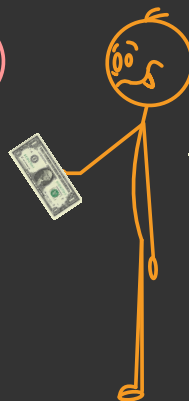
```
In [1]: 1.45 * 15  
Out[1]: 21.75
```



I want to exchange
\$40 for canadian money

here's **\$58**

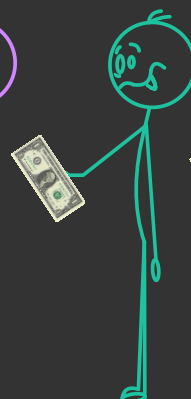
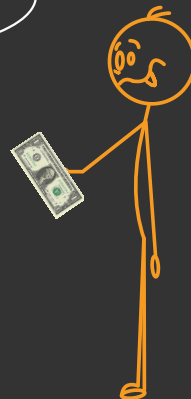
```
In [2]: 1.45 * 40  
Out[2]: 58.0
```



I want to exchange
\$32 for canadian money

here's **\$46.40**

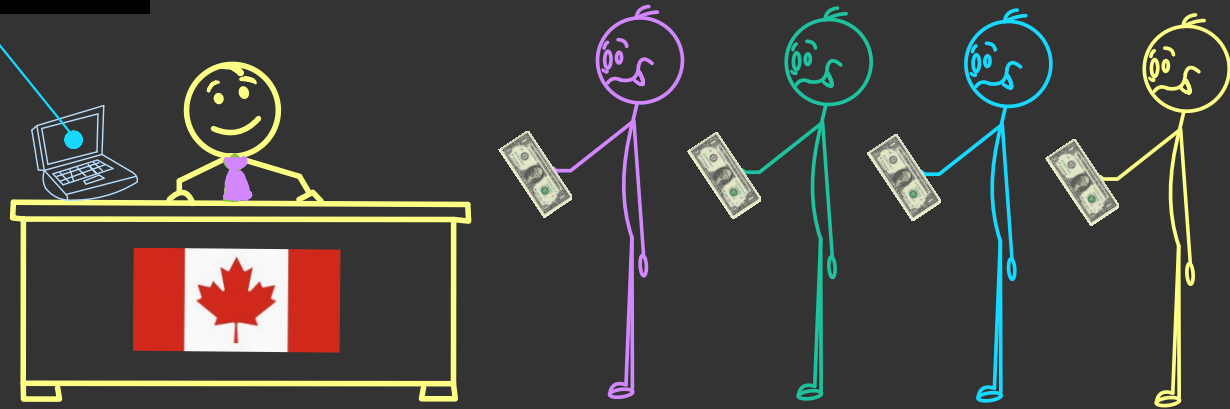
```
In [3]: 1.45 * 32  
Out[3]: 46.4
```



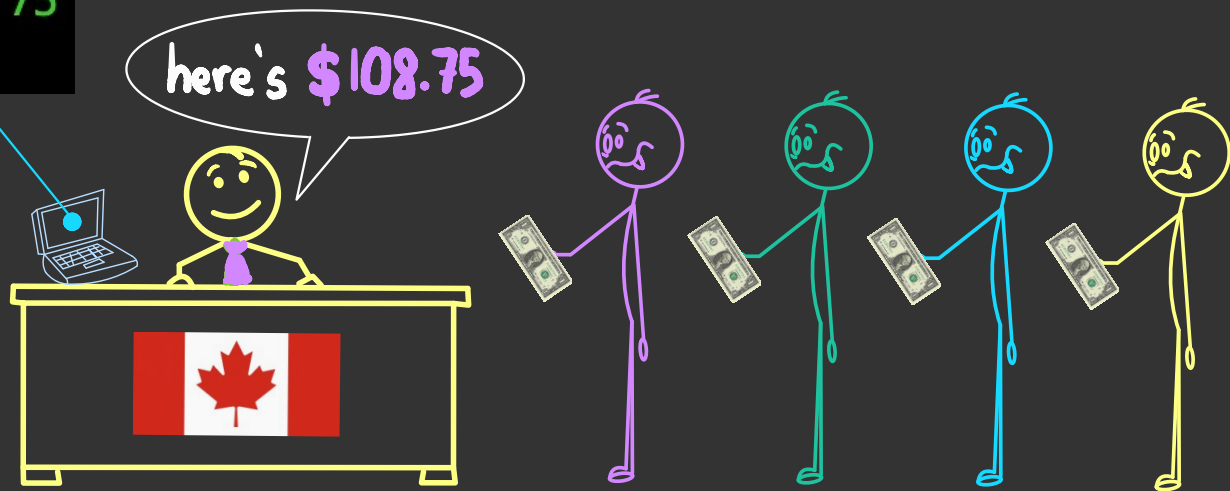


variables are useful when you need to use the same value repeatedly

```
In [4]: rate = 1.45
```



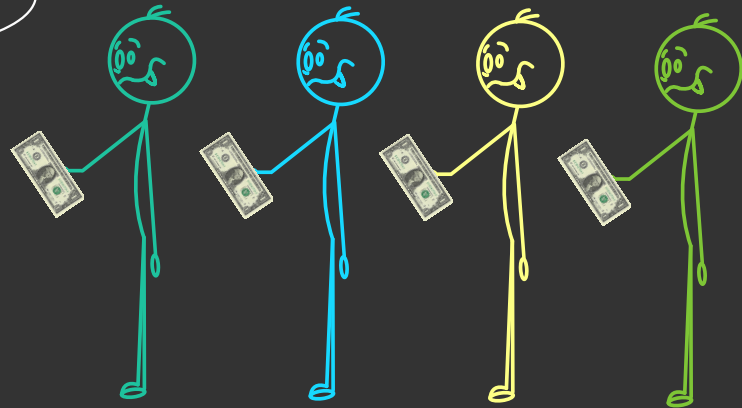
```
In [5]: rate * 75  
Out[5]: 108.75
```



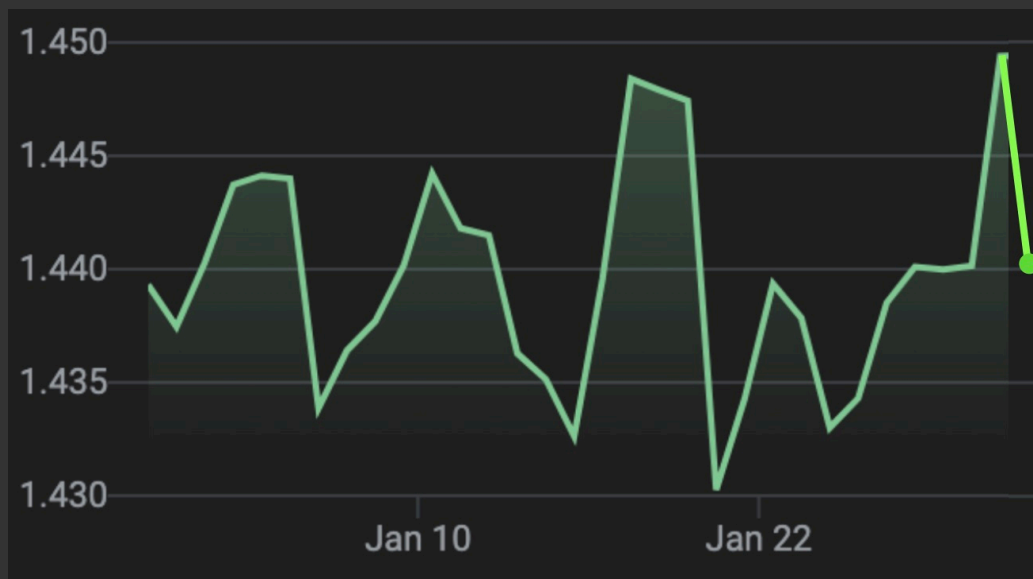
```
In [6]: rate * 25  
Out[6]: 36.25
```

I want to exchange
\$25 for canadian money

here's **\$36.25**

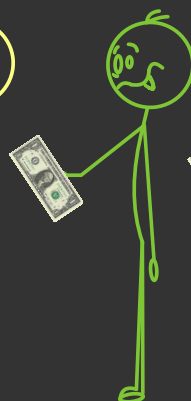
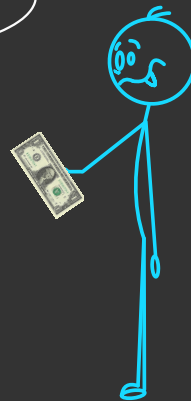


variables are also useful when the value is
subject to change



I want to exchange
\$50 for canadian money

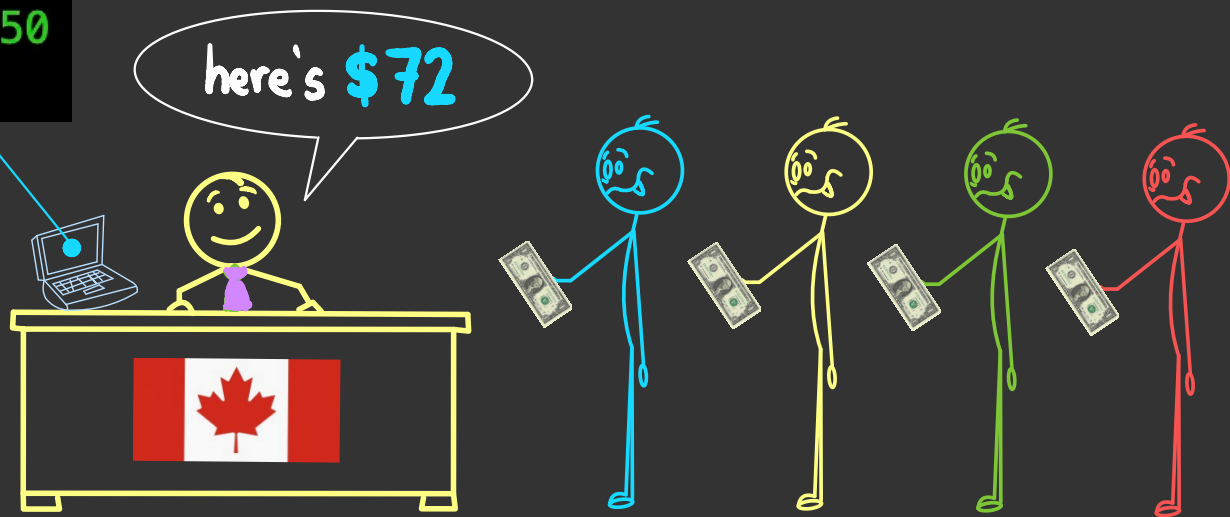
one moment
please



```
In [7]: rate = 1.44
```



```
In [8]: rate * 50  
Out[8]: 72.0
```



the assignment statement

variable

=

expression

a valid
variable name

- ▶ only letters, numbers, underscores
- ▶ cannot begin with a number

any expression is
allowed on the
right-hand side

the assignment statement

variable = expression

```
In [1]: rate = 1.35
```

```
In [2]: rate  
Out[2]: 1.35
```

```
In [3]: average = (10+11+12)/3
```

```
In [4]: average  
Out[4]: 11.0
```

```
In [5]: cdn = rate * 15
```

```
In [6]: cdn  
Out[6]: 20.25
```

```
In [7]: injury = "boo" * 2
```

```
In [8]: injury  
Out[8]: 'booboo'
```

```
In [9]: social_security_number = "yeahright"
```

```
In [10]: social_security_number  
Out[10]: 'yeahright'
```

the **assignment** statement
is not symmetric

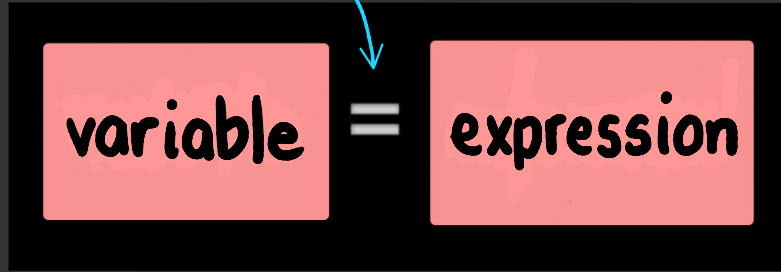
variable = expression



expression = variable



in fact, it's best not to see
this as an equals sign



the equivalent english statement is
"let **variable** be **expression**"

a currency exchange app

```
money = input("How many US dollars do you want to exchange? ")  
rate = 1.45  
print("The value of your currency in Canadian dollars:")  
print(rate * float(money))
```

Code

←
saved as

currency_exchange.py

- (base) \$ python currency_exchange.py
How many US dollars do you want to exchange? 15
The value of your currency in Canadian dollars:
21.75
- (base) \$ python currency_exchange.py
How many US dollars do you want to exchange? 40
The value of your currency in Canadian dollars:
58.0

Terminal

